

Historical- CHE 101: ‘Introduction to Chemical Engineering Concepts’

Current Description and Topics

Current Catalog Course Description:

Overview of engineering and chemical principles used in chemical engineering technology. Thermodynamics, transport phenomena, and reaction engineering applied to process and product design. Course Information: Prerequisite(s): Credit or concurrent registration in [MATH 180](#); and Credit or concurrent registration in [CHEM 122](#) and Credit or concurrent registration in [CHEM 123](#). **3 credit hours:**

Course Learning Outcomes from Course Request System (CRS):

This course is intended to (1) present the historical origins and paradigms of chemical engineering, (2) prepare freshmen for the intellectual culture of chemical engineering through concrete problems and demonstrations that preview 200 and 300 level ChE requirements, (3) develop an understanding of the technological, societal, environmental and regulatory context of chemical engineering, (4) expose connections of chemical engineering to basic sciences and other engineering disciplines, and (5) indicate the role of chemical engineers in a variety of traditional industries and emerging fields.

Current CHE 101 Major Topics

- Genesis of chemical engineering as a field, paradigm of unit operations, historical achievements of chemical engineering, what chemical engineers do in varied industries- 3 hours
- Relation of ChE to sciences and other engineering fields, and technological frontiers in chemical engineering- 3 hours
- Units and conversions, dimensional analysis and dimensionless groups, conservation principle, example process flowsheet- 3 hours
- Shear stress and viscosity, mechanical work done by fluids, Reynolds number and flow regimes, history of thermodynamics, steam engines- 3 hours
- Phase behavior, vapor pressure curve, power of steam (demonstration), heat capacity of air (demonstration and calculation), distillation columns and operating lines- 3 hours
- Hydrogen generation and storage, hydrogen engines and fuel cells- 3 hours
- Diffusion (random walk – simulation), mass transfer coefficient, fluid mixing- 2 hours
- Thermodynamics of reactions, Gibbs energy and equilibrium constant, extent of reaction and simple equilibrium calculation- 3 hours
- Reaction kinetics and simple rate laws, CSTR and PFR reactors- 3 hours
- Environmental pollution, regulatory issues, hydrology and contaminant transport (including animated simulation)- 3 hours
- Membranes, desalination, wastewater treatment, nanotechnology concepts- 3 hours
- Heat transfer by conduction and convection, Newton's law of cooling (demonstration and calculation), wind chill factor- 3 hours
- Biochemical engineering concepts and applications- 3 hours
- Computers, programming and math packages: Tank drainage problem worked with Fortran, C, Java, Matlab, Maple- 3 hours

Quizzes & Exams- 4 hours

45 hours Total

CHE 101 Course Curriculum Proposal

- Change title to “Introduction to Sustainable Engineering”
- Change pre-requisites
- Revise course topics
- Add course to core CHE BS degree program

Sustainable Engineering- Process Development

Processes that:

- Produce more energy than they consume
- Use renewable energy rather than fossil fuels
- Have a positive contribution to the environment
- Produce minimal waste, consume by-products
- Create products without need of further processing and/or recycling

Some Sustainable Technology Examples

- Renewable energy technologies
- Energy efficiency technologies
- Green transportation
- Water conservation and management
- Construction materials and design
- Waste management
- Agriculture
- Carbon capture and storage
- Green chemistry
- Clean energy storage

Sustainable Development in Chemical Process Design

- Goal of chemical process design sustainability:
 - Minimize resource use
 - Prevent or reduce releases
 - Increase economic feasibility of the process
 - Minimize, substitute, moderate, simplify
 - Prioritize the planet and people's safety

Ref: Chemical Processing May 3, 2023

A Few Sustainable Problems that Chemical Engineers can Solve

- Scale-up from novel lab-scale processes to industrial size processes
- Process optimization to be more economical, environmentally friendly, and energy efficient
- Reduce waste and shipping weight by concentrating products without compromising on quality.
- Replace petroleum-based products, with plant-based materials that perform equally as well or better
- Reduce microplastics

A Few Sustainable Problems that Chemical Engineers can Solve

- Improve energy efficiency, energy storage, and energy conversion for fuels and materials
- Improve carbon capture technologies and integrate them with other industries such as wastewater management.

Ref: [Chemengstudent.com/the-role-of-chemical-engineering-in-sustainable-development](https://chemengstudent.com/the-role-of-chemical-engineering-in-sustainable-development)